1. $x^2y'' + xy' - y = 0$

解：取 $y = x^r$

$$x^2 r(r-1)x^{r-2} + x \cdot r \cdot x^{r-1} - x^r = 0$$

$$\Rightarrow \quad r(r-1)x^r + rx^r - x^r = 0$$

$$\Rightarrow r^2 - 1 = 0 \Rightarrow r = \pm 1$$

$y = x$ 和 $y = \frac{1}{x}$ 是原方程的两个线性无关的解.

通解为 $y = C_1 x + C_2 \frac{1}{x}$

2. $y'' + xy' - y = 0$

$p(x) = x$, $Q(x) = -1$ 在 $x=0$ 处解析. $x=0$ 为正常点

$$y = \sum_{n=0}^{\infty} C_n x^n, \quad y' = \sum_{n=1}^{\infty} C_n n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} C_n n(n-1) x^{n-2}$$

$$= \sum_{n=0}^{\infty} (n+1) C_{n+1} x^n \qquad = \sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} x^n$$

代入原方程得:

$$\sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} x^n + \sum_{n=0}^{\infty} (n+1) C_{n+1} x^{n+1} - \sum_{n=0}^{\infty} C_n x^n = 0$$

由 $\sum_{n=0}^{\infty} (n+1) C_{n+1} x^{n+1} = \sum_{n=0}^{\infty} n C_n x^n$,

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1) C_{n+2} + n C_n - C_n \right] x^n$$

$$= \sum_{n=0}^{\infty} \left[(n+2)(n+1) C_{n+2} + (n-1) C_n \right] x^n = 0$$

对比系数得 $C_{n+2} = \dfrac{(1-n) C_n}{(n+2)(n+1)}$, $n = 0, 1, 2, \cdots$

$$C_2 = \frac{C_0}{2}, \ C_3 = 0, \ C_4 = \frac{-C_2}{4 \cdot 3} = \frac{-C_0}{4 \cdot 3 \cdot 2}, \ C_5 = \frac{-2C_3}{5 \cdot 4} = 0, \cdots$$

综上、方程的通解为

$$y=\sum_{n=0}^{\infty}C_nx^n=C_0+C_1x+\frac{C_0}{2}x^2-\frac{C_0}{24}x^4+\cdots$$

$$=C_0\left(1+\frac{1}{2}x^2-\frac{1}{24}x^4+\cdots\right)+C_1x$$

3、$xy''+\frac{1}{2}y'-y=0$

解：$x=0$ 为正则奇点

$$y=\sum_{n=0}^{\infty}C_nx^{n+r}$$

$$y'=\sum_{n=0}^{\infty}C_n(n+r)x^{n+r-1}$$

$$y''=\sum_{n=0}^{\infty}C_n(n+r)(n+r-1)x^{n+r-2}$$

$$xy''+\frac{1}{2}y'-y=\sum_{n=0}^{\infty}C_n(n+r)(n+r-1)x^{n+r-1}$$

$$+\frac{1}{2}\sum_{n=0}^{\infty}C_n(n+r)x^{n+r-1}-\sum_{n=0}^{\infty}C_nx^{n+r}$$

$$=C_0r(r-1)x^{r-1}+\frac{1}{2}C_0rx^{r-1}$$

$$+\sum_{n=1}^{\infty}\left[C_n(n+r)(n+r-1)+\frac{1}{2}C_n(n+r)-C_{n-1}\right]x^{n+r-1}=0$$

$$\Rightarrow \begin{cases} C_0 r(r-1) + \frac{1}{2} C_0 r = 0 \\ C_n (n+r)(n+r-1) + \frac{1}{2} C_n (n+r) - C_{n-1} = 0, \quad n=1,2,\dots \end{cases}$$

$$C_0\left(r^2 - \frac{1}{2} r\right) = 0 \Rightarrow r=0,\ r=\frac{1}{2}$$

$r=0$, $\quad C_n \cdot n(n-1) + \frac{1}{2} C_n \cdot n - C_{n-1} = 0. \quad n=1,2,\dots$

$$C_n = \frac{C_{n-1}}{(n - \frac{1}{2} n)}, \quad n=1,2,\dots$$

$$C_1 = 2C_0,\ \ C_2 = \underset{=2C_0}{C_1},\ \ C_3 = \frac{C_2}{3-\frac{3}{2}} = \frac{2}{3} C_2 = \frac{4}{3} C_0$$

$$C_4 = \frac{C_3}{2} = \frac{2}{3} C_0, \dots$$

$r=\frac{1}{2}$, $\quad C_n\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right) + \frac{1}{2} C_n\left(n+\frac{1}{2}\right) - C_{n-1} = 0$

$$C_n\left(n+\frac{1}{2}\right) \cdot n = C_{n-1} \Rightarrow C_n = \frac{C_{n-1}}{n^2 + \frac{1}{2} n}, \quad n=1,2,\dots$$

$$C_1 = \frac{2}{3} C_0,\ \ C_2 = \frac{C_1}{5} = \frac{2}{3\cdot 5} C_0,\ \ C_3 = \frac{C_2}{9+\frac{3}{2}} = \frac{2}{21} C_2 = \frac{4}{3\cdot 5\cdot 21} C_0$$

$\dots$

$y = C_1 y_1(x) + C_2 y_2(x)$. 其中

$$y_1 = C_0 + 2C_0 x + 2C_0 x^2 + \frac{4}{3} C_0 x^3 + \frac{2}{3} C_0 x^4 + \cdots$$

$$y_2 = C_0 x^{\frac{1}{2}} + \frac{2}{3} C_0 x^{\frac{3}{2}} + \frac{2}{15} C_0 x^{\frac{5}{2}} + \frac{4}{15\cdot 21} C_0 x^{\frac{7}{2}} + \cdots$$

4. $\begin{cases} \dfrac{dx}{dt} = -2x+y+2e^{-t} \\ \dfrac{dy}{dt} = x-2y \end{cases}$

令 $D=\dfrac{d}{dt}$

则 $(D+2)x-y=2e^{-t}$ (1)

$(D+2)y-x=0$ (2)

消去 y. (1)式乘 $(D+2)$

$(D+2)^2x-(D+2)y=2(D+2)e^{-t}=-2e^{-t}+4e^{-t}$ (3)

(3)+(2) 得 $(D+2)^2x-x=-2e^{-t}+4e^{-t}=2e^{-t}$

$(D^2+4D+3)x=2e^{-t}$ (4)

由 $m^2+4m+3=0 \Rightarrow m=-1, m=-3$

齐次解为 $C_1e^{-t}+C_2e^{-3t}$

设 $y_p=ate^{-t}$ 代入(4) 得

$a(e^{-t}t-2e^{-t})+4(a(-e^{-t}(t-1))+3ate^{-t}=2e^{-t}$

$ae^{-t}t-2ae^{-t}-4ate^{-t}+4ae^{-t}+3ate^{-t}=2e^{-t}$

$\Rightarrow 2ae^{-t} = 2e^{-t}$，$a=1$

$y_p = te^{-t}$ 即为(4)的特解

$x = C_1e^{-t} + C_2e^{-3t} + te^{-t}$ 为(4)通解.

由(1) $y = (D+2)x - 2e^{-t}$

$= -C_1e^{-t} - 3C_2e^{-3t} + e^{-t} - te^{-t}$

$+ 2C_1e^{-t} + 2C_2e^{-3t} + 2te^{-t} - 2e^{-t}$

$= C_1e^{-t} - C_2e^{-3t} + te^{-t} - e^{-t}$

$\Rightarrow$ 方程组通解：$x = C_1e^{-t} + C_2e^{-3t} + te^{-t}$

$y = C_1e^{-t} - C_2e^{-3t} + te^{-t} - e^{-t}$